

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Industrial and Systems Engineering

2. Subject Name: Optimization and Heuristics Methods Lab
Credits: 3

L-T-P: 0-0-3

3. Pre-requisites: Operations Research-I (IM 21201) & Operations Research-II (IM 21204)

4. Syllabus and reference books:

Syllabus:

Implementation of different heuristics and metaheuristics taught in the Optimization and Heuristics methods course in large size real world problems using software such as MATLAB, R and Python. Hands on Exercise, Mini Project. Hill Climbing Simulated Annealing, Tabu Search, Genetic Algorithm, Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Artificial Bees Colony (ABC), Greedy Heuristics, Multi-Objective Optimization, NSGA approach for Multi-Objective Optimization. Fuzzy Logic- Fuzzy Sets and Relation, Fuzzification and Defuzzification, Membership Functions.

Reference Books:

- 1) Handbook of Metaheuristics – Michel Gendreau and Jean-Yves Potvin
- 2) Multi-Objective Optimization using Evolutionary Algorithms – Kalyanmoy Deb
- 3) Fuzzy logic with Engineering applications - T J Ross.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Hill Climbing	1
2.	Simulated Annealing	1
3.	Tabu Search	1
4.	Genetic Algorithm	1
5.	Particle Swarm Optimization (PSO)	1
6.	Ant Colony Optimization (ACO)	1
7.	Artificial Bees Colony (ABC)	1
8.	Greedy Heuristics	1
9.	Multi-Objective Optimization	1
10.	NSGA approach for Multi-Objective Optimization	1
11.	Fuzzy Logic- Fuzzy Sets and Relation,	1

	Fuzzification and Defuzzification , Membership Functions.	
Total number of hours		33